Operational Challenges from the Commander Command Post

Managing vehicular flow within the Bengaluru metropolitan area requires navigating an exceptionally dense urban environment.[1] With a registered vehicle population exceeding 1.25 crore growing at an annual rate of 7% to 10%, contrasted against an annual human population growth rate of 3.7%, the city's infrastructure is continuously pushed beyond its structural limits.[1] This rapid vehicular growth operates across a road network of approximately 14,000 km, managed by 53 jurisdictional traffic police stations distributed across four geographic divisions.[3] The resulting congestion spans more than 1,638 km of daily gridlock, with closely spaced junctions operating far beyond their design capacities.[1]

For senior command staff and station house officers, the daily task of managing this gridlock is characterized by constant operational pressure.[3] The system is highly sensitive to disruptions: if an officer or traffic volunteer is distracted or steps away from a critical intersection for even a brief period, vehicular queues build rapidly, causing gridlock that can take hours to clear.[3] This operational strain is worsened by widespread traffic violations alongside the physical toll of working in high levels of air pollution and intense monsoon downpours.[3]

Exploratory Data Analysis (EDA) of the active operational logs reveals that the city's traffic disruptions are overwhelmingly dominated by "the mundane." More than 80% of recorded incidents are unplanned vehicle breakdowns occurring on non-corridor minor roads, whereas high-profile planned events (such as cricket matches at the M. Chinnaswamy Stadium, political rallies, and religious festivals) represent a small fraction of the operational database.[3] Crucially, the dataset is marked by extreme sparsity, with critical structural fields such as cargo classifications, breakdown reasons, vehicle age, route paths, and officer assignments having missing rates between 96.6% and 100%. To address these real-world data limitations within a 6-day prototype window, this framework pivots to a sparse-data systems architecture. This approach combines natural language processing (NLP), prescriptive heuristics, and robust statistical estimators to bypass missing fields while maintaining complete mathematical and system integrity.

Data Ingestion, Structural Mapping, and Sparsity Analysis

To build an operational prototype, we must first document the fields available and account for their high missing rates. The table below maps the 46 database columns, incorporating the actual dataset health to focus development only on populated fields:

Column Name	Data Type	Mathematical Symbol	Role in System Architecture and	Sparsity/Health Status

			Optimization Engines	
id	BigInt (PK)	ID_k	Unique database index; tracks and pairs specific historical records.	Healthy (0% Null)
event_type	Categorical	E_c	Categorizes events; determines base capacity scaling.[2]	Healthy (0% Null)
latitude	Decimal	lat_k^{start}	Starting latitude of the event or localized bottleneck boundary.	Healthy (0% Null)
longitude	Decimal	lon_k^{start}	Starting longitude of the event or localized bottleneck boundary.	Healthy (0% Null)
endlatitude	Decimal	lat_k^{end}	Ending latitude of the spatial bottleneck or work zone.	Sparsely Populated
endlongitude	Decimal	lon_k^{end}	Ending longitude of the spatial bottleneck or	Sparsely Populated

			work zone.	
address	VarChar	Add_k^{start}	Human-readable start address for spatial routing.[2]	Healthy (0% Null)
end_address	VarChar	Add_k^{end}	Human-readable end address; defines physical diversion zone.	Sparsely Populated
event_cause	Categorical	C_e	Identifies the physical cause of capacity reduction.[2]	Healthy (0% Null)
requires_road_closure	Boolean	$I_{closure}$	Binary trigger that zeroes out link capacity in routing.	Healthy (0% Null)
start_datetime	DateTime	T_k^{start}	Beginning of the temporal indicator function.	Healthy (0% Null)
end_datetime	DateTime	T_k^{end}	End of the temporal indicator function.	Severely Sparse (94.2% Null)
status	Categorical	S_k	Tracks state of the event (e.g., Active, Resolved).[2]	Healthy (0% Null)
authenticated	Boolean	A_k	Validation flag; filters out	Healthy (0%

			unverified citizen reports.[2]	Null)
modified_datetime	DateTime	T_k^{mod}	Last status modification timestamp; crucial proxy for resolution.	Healthy (0% Null)
map_file	VarChar	M_f	Path to shapefile or KML file mapping closure geometry.[2]	Sparsely Populated
direction	Categorical	D_a	Directional constraint on flow vectors; isolates lanes.	Sparsely Populated
description	Text	D_{text}	Messy bilingual text parsed to extract recovery risk.	Highly Populated (83.4%)
veh_type	Categorical	V_{type}	Identifies the vehicle class (e.g., HGV, Bus, Two-Wheeler).	Healthy (0% Null)
veh_no	VarChar	V_{reg}	Vehicle registration number used for towing log auditing.[2]	Sparsely Populated
corridor	Categorical	L_{corr}	Maps the incident directly to a	Sparsely Populated

			major arterial corridor.[2]	
priority	Integer	P_r	Dynamic priority score; coordinates agency dispatch.	Healthy (0% Null)
cargo_material	Categorical	M_{cargo}	Determined physical towing risk; replaced by text mining.	Severely Sparse (96.6% Null)
reason_breakdown	Categorical	R_{break}	Determined towing clearance difficulty; replaced by text mining.	Severely Sparse (96.6% Null)
age_of_truck	Integer	Age_{veh}	Determined asset age; replaced by text mining.	Severely Sparse (96.6% Null)
created_date	DateTime	T_k^{create}	Baseline timestamp for initial ticket generation.	Healthy (0% Null)
route_path	Text (JSON)	P_{route}	List of coordinate nodes; replaced by User Equilibrium routing.	Severely Sparse (98.3% Null)

client_id	Integer	Cl_{id}	Tracks system clients or agencies accessing the database.[2]	Healthy (0% Null)
created_by_id	Integer	Usr^{create}	Identifier of the operator or automated sensor logging incident.	Healthy (0% Null)
last_modified_by_id	Integer	Usr^{mod}	Identifier of the operator performing status modification.	Healthy (0% Null)
assigned_to_police_id	Integer	Pol_{id}	ID of assigned officer; replaced by prescriptive optimization.	Severely Sparse (98.4% Null)
citizen_accident_id	BigInt	Cit_{acc}	Foreign key linking to public safety database.[2]	Sparsely Populated
comment	Text	C_{text}	Commander's notes; replaced by description column parsing.	Severely Sparse (100% Null)
police_station	Categorical	Ω_p	Jurisdictional boundary; constrains optimization scheduling.	Healthy (0% Null)

meta_data	Text (JSON)	$Meta_{json}$	Semi-structured telemetry; replaced by description parsing.	Severely Sparse (100% Null)
kgid	VarChar	$KGID$	Karnataka Government ID of the responding police officer.	Healthy (0% Null)
resolved_at_address	VarChar	Add^{res}	Physical address where the incident was officially resolved.	Sparsely Populated
resolved_at_latitude	Decimal	lat^{res}	Spatial latitude of the final resolution point.	Sparsely Populated
resolved_at_longitude	Decimal	lon^{res}	Spatial longitude of the final resolution point.	Sparsely Populated
closed_by_id	Integer	Usr^{close}	ID of the station officer verifying and closing the ticket.	Healthy (0% Null)
closed_datetime	DateTime	T_k^{close}	Timestamp marking administrative closure; contains heavy	Highly Populated (38.5%)

			lag.	
resolved_by_id	Integer	Usr^{res}	ID of the officer or operator who completed clearance.	Sparsely Populated
resolved_datetime	DateTime	T_k^{res}	Timestamp when physical flow was restored; contains high lag.	Sparsely Populated
gba_identifier	VarChar	GBA_k	Geographic Block Area identifier for spatial clustering.	Sparsely Populated
zone	Categorical	Z_k	Division-level zoning index (e.g., East, West, South, North).	Healthy (0% Null)
junction	Categorical	v	Target intersection node impacted by queue spillback.[6]	Healthy (0% Null)

Global Precedents and Comparative Methodological Review

To address event-driven congestion effectively, cities have historically deployed adaptive signal control systems, dynamic traffic routing, and simulation-based planning tools. Singapore's Land Transport Authority manages congestion through the Expressway Monitoring and Advisory System (EMAS), which combines real-time incident detection with dynamic variable message signs to guide motorists. London's SCOOT (Split Cycle Offset Optimisation Technique) uses real-time inductive loop telemetry to optimize signal timing dynamically.[1] In New York City, the Midtown in Motion active traffic management program dynamically adjusts arterial signals to handle heavy traffic and reduce gridlock at highly oversaturated intersections.[1]

For predictive modeling and real-time operations, Perth and Oxfordshire deployed Aimsun Predict within the Future Road Lab. This system combines a Random Forest model for long-term day patterns with an online learning algorithm. By using a modular framework that updates specific sub-models when prediction accuracy drifts, Aimsun Predict maintains low forecasting errors (measured by the percentage of GEH under 5, i.e., $\%GEH < 5$) even during major, structural disruptions.

In India, Kolkata's traffic department addressed patrol scheduling by implementing a Fuzzy Goal Programming model combined with a Genetic Algorithm. This approach optimizes the deployment of patrol vehicles across shifts, balancing cost constraints against the need for enforcement at high-risk, accident-prone intersections. Additionally, research on the Tarrant County network in North Texas demonstrated the value of Bilevel Optimization models. These models minimize network-wide travel times during lane closures by factoring in driver response patterns under User Equilibrium constraints.

The critical lesson from these global precedents is that they all adapt to imperfect telemetry. When on-field sensors or database records degrade, successful systems pivot to robust mathematical estimators and prescriptive optimization rather than relying on perfect historical logs.

Redefining Breakdown Clearance Duration (τ_c) Under Sparsity

Because cargo_material, reason_breakdown, and age_of_truck are 96.6% NULL, and comment and meta_data are 100% NULL, the previous formulation for clearance time cannot be calculated. However, the unstructured description column is 83.4% populated and contains critical, raw descriptions in mixed Kannada and English (e.g., *"KSRTC bus breakdown near Silk Board due to axle axle murida raste madhya"* or *"heavy water logging and commercial vehicle puncture"*).

We redefine the clearance duration τ_c using an Accelerated Failure Time (AFT) survival model. AFT models are statistically robust to heavy-tailed incident data and account for censored observations. The model relies only on healthy fields: vehicle type (veh_type), priority (priority), spatial coordinates, and semantic features extracted from the messy text.

Bilingual Feature Extraction via LLM

We process the free-text description column (D_{text}) using a lightweight cross-lingual transformer model (such as XLM-RoBERTa or mBERT fine-tuned for code-mixed Indian languages) to handle multilingual code-mixing. The model maps the text into a dense representation:

$$\mathbf{h}_{LLM} = \mathrm{Encoder}(D_{text}) \in \mathbb{R}^d$$

From this representation, we compute a high-risk semantic clearance indicator $S_{risk} \in $ using a linear projection with sigmoid activation:

$$S_{risk} = \frac{1}{1 + \exp(-\mathbf{w}^T \mathbf{h}_{LLM})}$$

The weight vector $\mathbf{w}$ is trained on a synthetic subset of annotated descriptions to classify high-impact events (e.g., heavy towing required, multi-vehicle collision, engine fire, or deep waterlogging).

Mathematical Formulation of Redefined τ_c

The redefined AFT survival model formulation (assuming a Weibull baseline hazard) is:

$$\ln(\tau_c) = \beta_0 + \boldsymbol{\beta}_{veh} \cdot \mathbf{v}_{type} + \beta_p \cdot P_r + \beta_d \cdot d_{depot}(lat, lon) + \beta_s \cdot S_{risk} + \sigma \cdot W$$

Taking the exponent, we obtain the predictive formula for clearance duration:

$$\tau_c = \exp(\beta_0 + \boldsymbol{\beta}_{veh} \cdot \mathbf{v}_{type} + \beta_p \cdot P_r + \beta_d \cdot d_{depot}(lat, lon) + \beta_s \cdot S_{risk} + \sigma \cdot W)$$

where:

- $\mathbf{v}_{type}$ is the one-hot encoded vector of the categorical veh_type column (e.g., HGV, Bus, LMV, Two-Wheeler).
- $P_r \in \{1, 2, 3\}$ is the integer value from the priority column.
- $d_{depot}(lat, lon)$ is the calculated Euclidean distance from the incident coordinates (latitude, longitude) to the nearest police towing or emergency recovery depot, acting as a proxy for physical response dispatch lag.
- S_{risk} is the LLM-extracted bilingual text risk score.
- W is the error term following the standard Gumbel (extreme value) distribution, which models the heavy-tailed tail of long-duration towing operations.
- σ is the Weibull scale parameter.

By training this model using maximum likelihood estimation (MLE) on the 38.5% of cases that have valid resolution proxies, we obtain a robust, text-enhanced predictor that operates under extreme tabular sparsity.

Zero-Shot Multi-Objective Resource Allocation

Optimization

Because assigned_to_police_id is 98.4% NULL, there is no historical training target for police deployment. To resolve this, we pivot to a **Zero-Shot Prescriptive MOBLP** formulation. Instead of learning from past behavior, we construct a prescriptive constraint matrix and define the baseline "Demand Coverage" (D_{ij}) using traffic flow theory, roadway capacity, and spatial risk.[6]

Prescriptive Calculation of Baseline "Demand Coverage" (D_{ij})

We define the baseline manpower demand D_{ij} (representing the minimum number of officers required at junction i during shift j) using a prescriptive formulation [6]:

$$D_{ij} = \max\left(D_i^{min}, \lceil w_1 \cdot \text{Class}_i + w_2 \cdot M_j(t) + w_3 \cdot H_{ij} \rceil\right)$$

where:

- $D_i^{min} \in \{1, 2\}$ is the absolute physical minimum requirement of personnel for junction safety.[6]
- $\text{Class}_i \in \{1, 2, 3\}$ is the junction size category, extracted programmatically by calculating the node degree (number of intersecting legs) from open-source maps (e.g., 5-leg arterial junctions are assigned Class 3, whereas minor 3-leg intersections are assigned Class 1).[1]
- $M_j(t)$ is the temporal peak factor, derived from start_datetime (T_k^{start}). It is set to 1.5 during morning peak (08:00 - 11:00) and evening peak (17:00 - 20:00) hours, and 1.0 otherwise.[6]
- H_{ij} is the dynamic spatiotemporal incident density, calculated as the count of active unplanned breakdowns occurring within a spatial radius $R = 1.5$ km of junction i during shift j [5]:

$$H_{ij} = \sum_{k \in \mathcal{N}_i} I_{active,k}(j)$$

- w_1, w_2, w_3 are prescriptive weighting coefficients calibrated in coordination with BTP station house officers (e.g., $w_1 = 0.5$, $w_2 = 1.0$, $w_3 = 0.8$).[6]

Heuristic-Based Constraints Matrix for NSGA-II

To optimize the binary decision variable $x_{ijk} \in \{0, 1\}$ (where $x_{ijk} = 1$ if officer k is deployed to junction i during shift j, and 0 otherwise) without historical training data, we construct a hard-coded heuristic constraint matrix [7]:

1. **Junction Demand Satisfaction:**

$$\sum_{k=1}^{N_c} x_{ijk} \geq D_{ij} \quad \forall i, j$$

2. **Single-Assignment Conflict Resolution:** An officer cannot be assigned to more than one junction during a given shift [6]:

$$\sum_{i=1}^{m} x_{ijk} \leq 1 \quad \forall j, k$$

3. **Jurisdictional Station Boundaries:** To ensure officers operate within their local commands, we group junctions by the police_station column (Ω_p) [3]:

$$\sum_{i \in \Omega_p} x_{ijk} \leq N_{p,k} \quad \forall j$$

   where $N_{p,k}$ is the total active strength of category k registered at station p.

4. **Spatiotemporal Feasibility Transition Constraint:** To prevent the genetic algorithm from generating physically impossible schedules (since route learning data is missing), we implement a proximity-based transition constraint [6]:

$$x_{ijk} \cdot x_{l,j+1,k} \cdot \Theta\left(d_{il} - D_{threshold}\right) = 0 \quad \forall i, l, j, k$$

   where:

- d_{il} is the pairwise Euclidean distance between junction i and junction l.
- $D_{threshold}$ is the maximum distance an officer can travel during a shift transition (e.g., 2.0 km).[5]
- $\Theta(z)$ is the Heaviside step function:

$$\Theta(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise} \end{cases}$$

This constraint strictly zero-values the fitness of chromosomes that attempt to dispatch an officer to a distant intersection in consecutive shifts, ensuring operational feasibility.[6]

Proxy Timestamps and Mathematically Sound Actual Resolution Time

Because the target variable end_datetime is 94.2% NULL and closed_datetime is 61.5% NULL, we must establish a mathematically sound proxy for the "Actual Resolution Time" ($\hat{T}_k^{res}$) to calibrate our models and run the GEH closed-loop feedback loop.

Simply using closed_datetime introduces massive administrative lag, as police officers routinely leave tickets open for days or even weeks after the physical breakdown has been cleared.

Outlier Detection via Median Absolute Deviation (MAD)

We first calculate the raw recorded duration for each ticket k:

$$\Delta T_k^{raw} = T_k^{close\_or\_mod} - T_k^{start}$$

where:

- T_k^{start} is start_datetime.
- $T_k^{close\_or\_mod} = T_k^{close}$ if closed_datetime is populated; otherwise, $T_k^{close\_or\_mod} = T_k^{mod}$ (modified_datetime).

Because administrative lag introduces extreme, heavy-tailed outliers, we apply Median Absolute Deviation (MAD) anomaly detection grouped by incident class (defined by the combination of event_cause and veh_type). The MAD is highly robust to extreme values compared to standard deviation.

The median duration for a specific incident class is denoted as $\tilde{D}_{class}$. The median absolute deviation is calculated as:

$$MAD_{class} = \text{median}\left(| \Delta T_k^{raw} - \tilde{D}_{class} | \right)$$

We establish the maximum feasible physical clearance threshold (TH_{class}):

$$TH_{class} = \tilde{D}_{class} + c \cdot 1.4826 \cdot MAD_{class}$$

where c is a conservative scale factor (set to 3.0) and 1.4826 is the scaling factor to approximate a standard normal distribution. Any ticket where $\Delta T_k^{raw} > TH_{class}$ is flagged as an administrative lag outlier.

Fallback Logic for Actual Resolution Time ($\hat{T}_k^{res}$)

To reconstruct the actual physical resolution timestamp and bypass administrative lag, we implement a multi-stage fallback logic:

$$\hat{T}_k^{res} = \begin{cases} T_k^{mod} & \text{if } \Delta T_k^{raw} > TH_{class} \text{ and } T_k^{mod} - T_k^{start} \leq TH_{class} \\ T_k^{start} + \tau_c & \text{if } \Delta T_k^{raw} > TH_{class} \text{ and } T_k^{mod} - T_k^{start} > TH_{class} \\ \min\left(T_k^{close}, T_k^{mod}\right) & \text{otherwise} \end{cases}$$

1. **Case 1 (Standard Outlier Recovery):** If the raw duration exceeds the threshold, but the last database modification (modified_datetime) occurred within the feasible threshold, we use T_k^{mod}. This captures the moment the officer updated the ticket status on-field (e.g., typing "cleared" on the ASTraM BOT), even if the formal administrative closure of the ticket was delayed.
2. **Case 2 (Extreme Outlier Fallback):** If both the closure and the modified timestamps exceed the threshold (e.g., a ticket left open for 31 days with no interim modifications), we use the physical starting timestamp plus the clearance time τ_c predicted by our redefined survival AFT model.
3. **Case 3 (In-Bounds Clean Data):** If the raw duration is within the feasible threshold, we use the minimum of the closure and modification timestamps.

This proxy resolves the temporal target issue, providing a clean dataset to evaluate modeled flows against observed flows using the GEH statistic.

Actionable Recommendations for Hackathon Implementation

To deliver a winning, fully functional prototype within the 6-day timeline, we structure the engineering tasks as follows:

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+-----------------------------------------------------------------------------+
|                    6-DAY PROTOTYPE ROADMAP
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[redacted]

Day 1: Text Feature Extraction and Timestamp Cleaning

- **Task 1.1:** Write a Python preprocessing script to clean the description column. Use bilingual regular expressions to extract key terms in both English and Kannada (e.g., *"axle"*, *"puncture"*, *"breakdown"*, *"accident"*, *"water logging"*, *"towing"*, *"bussu"*, *"rasthe"*).
- **Task 1.2:** Load a lightweight pre-trained transformer (e.g., sentence-transformers/LaBSE or a multilingual BERT) to generate text embeddings. Train a simple classifier on a small, manually labeled subset of 200 rows to output the semantic risk score S_{risk}.
- **Task 1.3:** Calculate the raw duration ΔT^{raw} for all records. Group the dataset by event_cause and veh_type, compute the median and MAD for each group, and apply our fallback logic to populate the Actual Resolution Time ($\hat{T}_k^{res}$) for every incident.

Day 2: Predictor and Dynamic Routing Implementation

- **Task 2.1:** Fit the redefined Accelerated Failure Time (AFT) Weibull model using the lifelines library in Python. Treat the resolved incidents (where ΔT^{raw} is within the MAD threshold) as event-observed, and the outstanding/outlier tickets as censored, ensuring unbiased parameter estimation.

- **Task 2.2:** Parse the spatial nodes (junction, zone) and links (corridor, coordinates) to construct a directed graph $G = (V, A)$ using NetworkX. Map the starting coordinates (latitude, longitude) to their nearest network nodes.

Day 3: Zero-Shot Optimization Engine Development

- **Task 3.1:** Code the prescriptive Demand Coverage (D_{ij}) calculation. Use spatial indexing (e.g., scipy.spatial.KDTree) to quickly query active breakdowns within 1.5 km of each junction i at any shift j.[5]
- **Task 3.2:** Implement the NSGA-II genetic algorithm using the DEAP library in Python. Define the chromosome as a binary matrix $\mathbf{X}$ representing shift assignments.[6] Customize the mutation and crossover operators to enforce our proximity-based transition constraint, penalizing or repairing chromosomes that violate transition thresholds.[6]

Day 4: Bilevel Diversion and Barricading Engine

- **Task 4.1:** Write a lightweight Frank-Wolfe algorithm in NumPy to solve the lower-level User Equilibrium traffic assignment on the network graph G.
- **Task 4.2:** Implement the upper-level simulated annealing algorithm to select the optimal links $a \in A$ to place physical barricades ($y_a = 1$). The objective function minimizes the total system travel time output by the Frank-Wolfe simulator.

Day 5: Closed-Loop Validation and GEH Integration

- **Task 5.1:** Use the resolved proxy timestamps ($\hat{T}_k^{res}$) to reconstruct hourly traffic volumes on the network.
- **Task 5.2:** Program the GEH statistic calculation for each monitored link:

$$GEH_a = \sqrt{\frac{2 \cdot (V_{sim,a} - V_{obs,a})^2}{V_{sim,a} + V_{obs,a}}}$$

- **Task 5.3:** Set up the feedback trigger: if the percentage of links with $GEH < 5$ drops below 85% during peak hours, write a script to dynamically adjust the capacity reduction coefficients (β_a) in the simulation model, simulating an automated closed-loop learning system.

Day 6: Command Dashboard Integration and Packaging

- **Task 6.1:** Build a Streamlit web application for the Traffic Management Center command staff.
- **Task 6.2:** The dashboard must allow senior officers to:
1. Input an upcoming event's parameters (type, duration, location).
2. View the predicted congestion boundaries and backward-propagating queue shockwaves.
3. Visualize the NSGA-II optimized manpower allocation schedules mapped directly to kgid and police_station jurisdictions.
4. Display the recommended barricading and diversion coordinates on an interactive map.
5. Inspect the current data quality metrics, highlighting administrative lag outliers.
- **Task 6.3:** Package the code with a clear README.md and Dockerize the application to ensure it runs seamlessly for the judges.

This revised plan addresses the real-world limitations of the BTP dataset. By pivoting from idealized data assumptions to robust, sparse-data systems engineering, your prototype will demonstrate the deep, practical problem-solving that separates winners from average competitors.

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